

RTOS INTERRUPT HANDLING INTERVIEW QUESTIONS

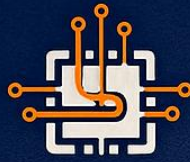


EmbeddedShiksha

RTOS Interrupt Interview Questions

1. What is an interrupt and why is it used in embedded systems?
2. Explain the interrupt lifecycle — what happens from the moment an interrupt occurs until the CPU returns to normal execution.
3. What is an ISR? What rules must you follow while writing an ISR?
4. Why must ISRs be short and deterministic?
5. What is interrupt latency? What factors affect it?
6. What is interrupt jitter?
7. What is the difference between maskable and non-maskable interrupts (NMI)?
8. What happens if an interrupt triggers while interrupts are disabled?
9. What is the difference between polling vs interrupt-based design?
10. How does an interrupt vector table work?
11. What are nested interrupts?
12. How does NVIC manage interrupt priority and preemption?
13. Who decides interrupt priorities: CPU, NVIC, or programmer?
14. What is preemption priority vs subpriority (priority grouping)?
15. What happens if a lower-priority interrupt is running and a higher-priority interrupt occurs?
16. Can a low-priority interrupt preempt a high-priority interrupt? Why or why not?
17. What is tail-chaining in Cortex-M processors?
18. What problems can arise from improper priority configuration?
19. What is interrupt starvation?
20. What is the purpose of disabling interrupts inside an ISR?
21. What is deferred interrupt handling?
22. Why do we need deferred execution instead of doing everything inside the ISR?
23. What is top-half and bottom-half interrupt architecture?
24. Give examples of deferred interrupt mechanisms in RTOS.

25. Explain with an example how a queue is used for deferred interrupt handling.
26. What is the difference between `xSemaphoreGiveFromISR()` and `xSemaphoreGive()`?
27. Why do we use `xHigherPriorityTaskWoken` in ISR-based APIs?
28. What happens if you don't call `portYIELD_FROM_ISR()`?
29. Can an ISR wake a higher priority task?
30. What happens if multiple interrupts try to defer work to the same queue?
31. What are the different ways an ISR can communicate with a task?
32. When should you use a queue vs semaphore for ISR-task communication?
33. Why are task notifications faster than semaphores?
34. Write a code snippet where an ISR notifies a task using task notification.
35. Explain how event groups can be used to signal multiple events from ISR.
36. How does FreeRTOS ensure safe API usage from ISR context?
37. Why can't you call blocking APIs like `vTaskDelay()` inside an ISR?
38. How do you avoid race conditions between ISR and task?
39. What happens if a queue is full when ISR tries to send data?
40. How does FreeRTOS handle priority inheritance in ISR-task sync?
41. If two interrupts occur at the same time, which one will run first?
42. If an ISR posts data to a queue and three tasks are waiting, which task gets it?
43. If ISR notifies a task but notification is missed, how do you debug the issue?
44. What will happen if ISR runs too long?
45. What design patterns do you follow to reduce interrupt load?
46. How do you guarantee real-time deadlines when interrupts are heavy?
47. What happens if interrupt priority is misconfigured in NVIC?
48. How do you measure interrupt latency in real hardware?
49. Can an ISR communicate with multiple tasks? How?
50. What will happen if an ISR triggers faster than the task can process? How do you fix it?



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